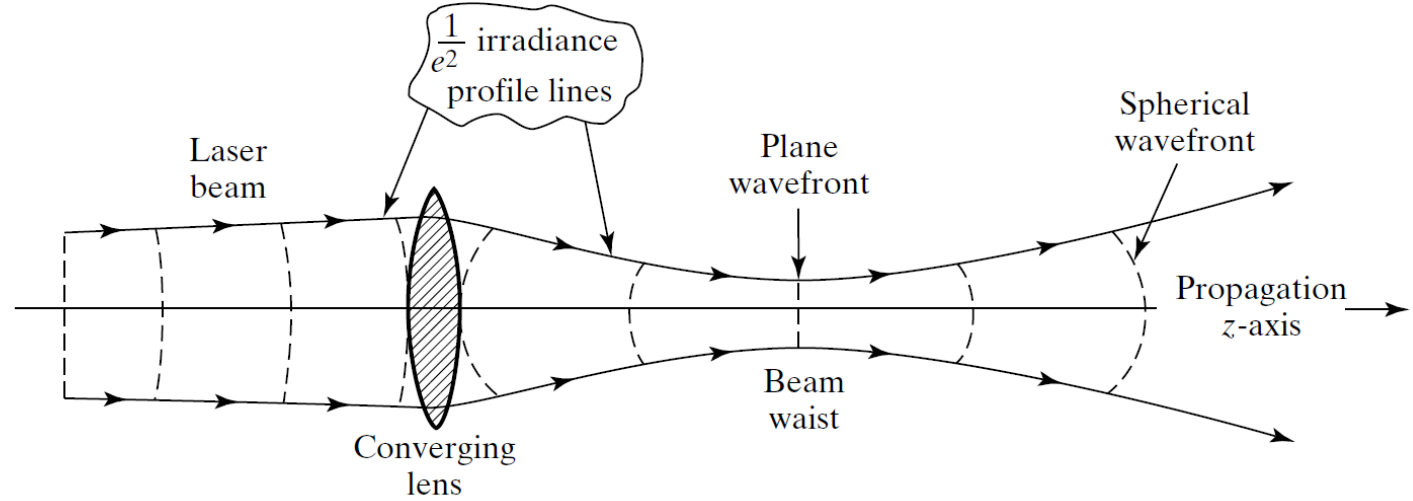


Gaussian Laser Beams

Figure 1 An external laser beam, confined essentially to regions within the $1/e^2$ -irradiance guidelines, is focused by a converging lens. The incoming beam is highly collimated so that its wavefronts are very nearly planar. The converging lens reshapes the wavefronts and focuses the beam—forming a beam waist—to the right of the lens. The beam diverges strongly as it propagates on past the beam waist.



$$\tilde{E}(x, y, z, t) = U(x, y, z)e^{i(kz - \omega t + \phi)} \quad (6)$$

The term $U(x, y, z)$ describes a departure from a “pure” plane wave and, when determined, provides the details that accurately specify the irradiance and phase variations of the wave. The exponential term in Eq. (6) merely reflects the “more-or-less” plane wave nature of the solution.

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + 2ik \frac{\partial U}{\partial z} = 0 \quad (9)$$

Equation (9) is a nontrivial partial differential equation with complex terms. To solve it, we make an “educated” guess¹ at a solution, motivated in part by the cylindrical symmetry (about the propagation direction) that we expect in the electric field and in part by the complex nature that the solution $U(x, y, z)$ must exhibit. Thus we “guess” the form

$$U(x, y, z) = E_0 e^{i\{p(z) + [k(x^2 + y^2)/2q(z)]\}} = E_0 e^{i\{p(z) + [k\rho^2/2q(z)]\}} \quad (10)$$

In the last equality we have introduced the cylindrical coordinate $\rho = \sqrt{x^2 + y^2}$.

$$E(x, y, z) = E_0 \frac{e^{ikz} e^{i\frac{k}{2z}(x^2 + y^2)}}{\sqrt{1 + \left(\frac{2z}{kw_0^2}\right)^2}} e^{-\frac{(x^2 + y^2)}{1 + \left(\frac{2z}{kw_0^2}\right)^2} \left(\frac{1}{w_0^2} + i\frac{k}{2z}\right)} e^{-i \tan^{-1} \frac{2z}{kw_0^2}}$$

Gaussian Laser Beams

$$E(x, y, z) = E_0 \frac{e^{ikz} e^{i \frac{k}{2z} (x^2 + y^2)}}{\sqrt{1 + \left(\frac{2z}{kw_0^2}\right)^2}} e^{-\frac{(x^2 + y^2)}{1 + \left(\frac{2z}{kw_0^2}\right)^2} \left(\frac{1}{w_0^2} + i \frac{k}{2z}\right)} e^{-i \tan^{-1} \frac{2z}{kw_0^2}}$$

$$E(\rho, z) = E_0 \frac{w_0}{w(z)} e^{-\frac{\rho^2}{w^2(z)}} e^{ikz + i \frac{k\rho^2}{2R(z)} - i \tan^{-1} \frac{z}{z_0}}$$

$$\rho^2 \equiv x^2 + y^2,$$

$$w(z) \equiv w_0 \sqrt{1 + z^2/z_0^2},$$

$$R(z) \equiv z + z_0^2/z,$$

$$z_0 \equiv \frac{kw_0^2}{2}$$

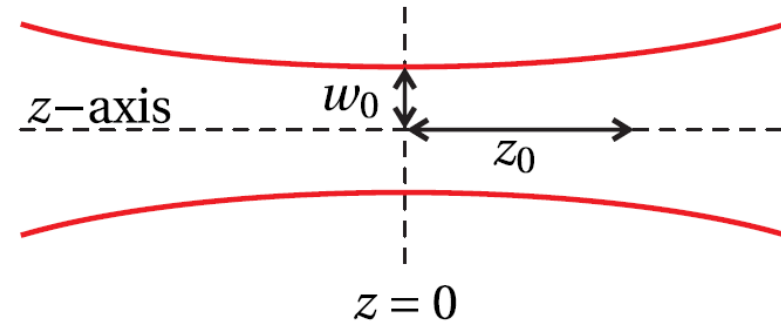


Figure 11.16 A Gaussian laser field profile in the vicinity of its beam waist.

Gaussian Laser Beams

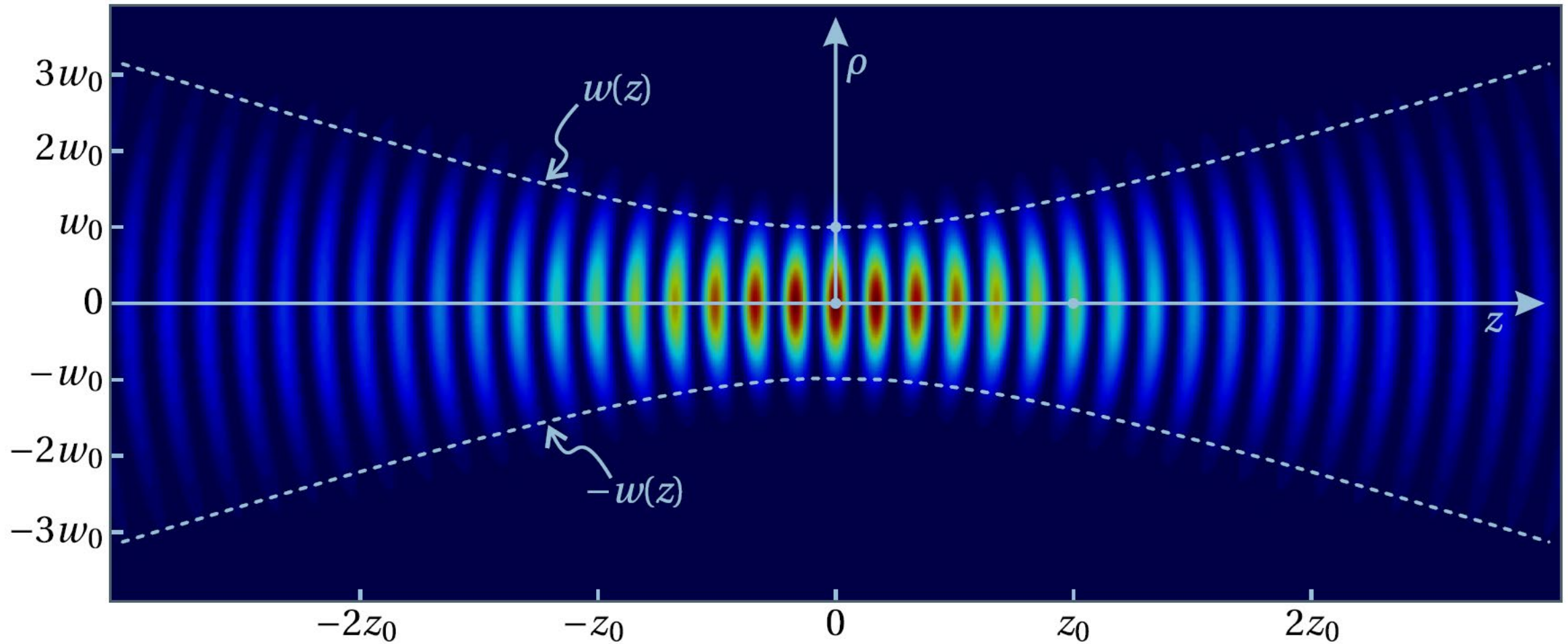


Figure 11.17 Real part of a Gaussian laser field at an instant in time. The radius of curvature of wavefronts is apparent.

Gaussian Laser Beams

$$\rho^2 \equiv x^2 + y^2,$$

$$w(z) \equiv w_0 \sqrt{1 + z^2/z_0^2},$$

$$R(z) \equiv z + z_0^2/z,$$

$$z_0 \equiv \frac{kw_0^2}{2}$$

intensity profile $I \propto |E|^2$

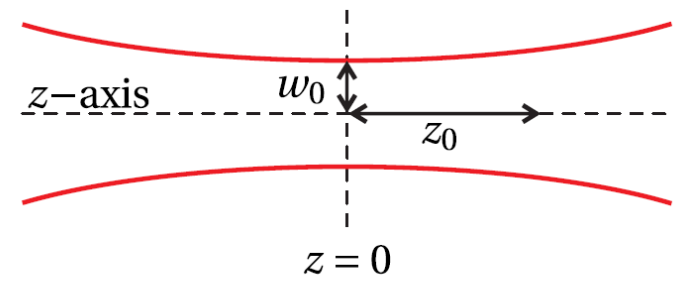


Figure 11.16 A Gaussian laser field profile in the vicinity of its beam waist.

$$I(\rho, z) = I_0 \frac{w_0^2}{w^2(z)} e^{-\frac{2\rho^2}{w^2(z)}} = \frac{I_0}{1 + z^2/z_0^2} e^{-\frac{2\rho^2}{w^2(z)}}$$

By inspection, we see that $w(z)$ gives the radius of the beam anywhere along z . At $z = 0$, the *beam waist*, $w(z = 0)$ reduces to w_0 , as expected. The parameter z_0 , known as the *Rayleigh range*, specifies the distance along the axis from $z = 0$ to the point where the intensity decreases by a factor of 2. Note that w_0 and z_0 are not independent of each other but are connected through the wavelength according to (11.44). There is a tradeoff: a small beam waist means a short *depth of focus*. That is, a small w_0 means a small Rayleigh range z_0 .

Gaussian Laser Beams

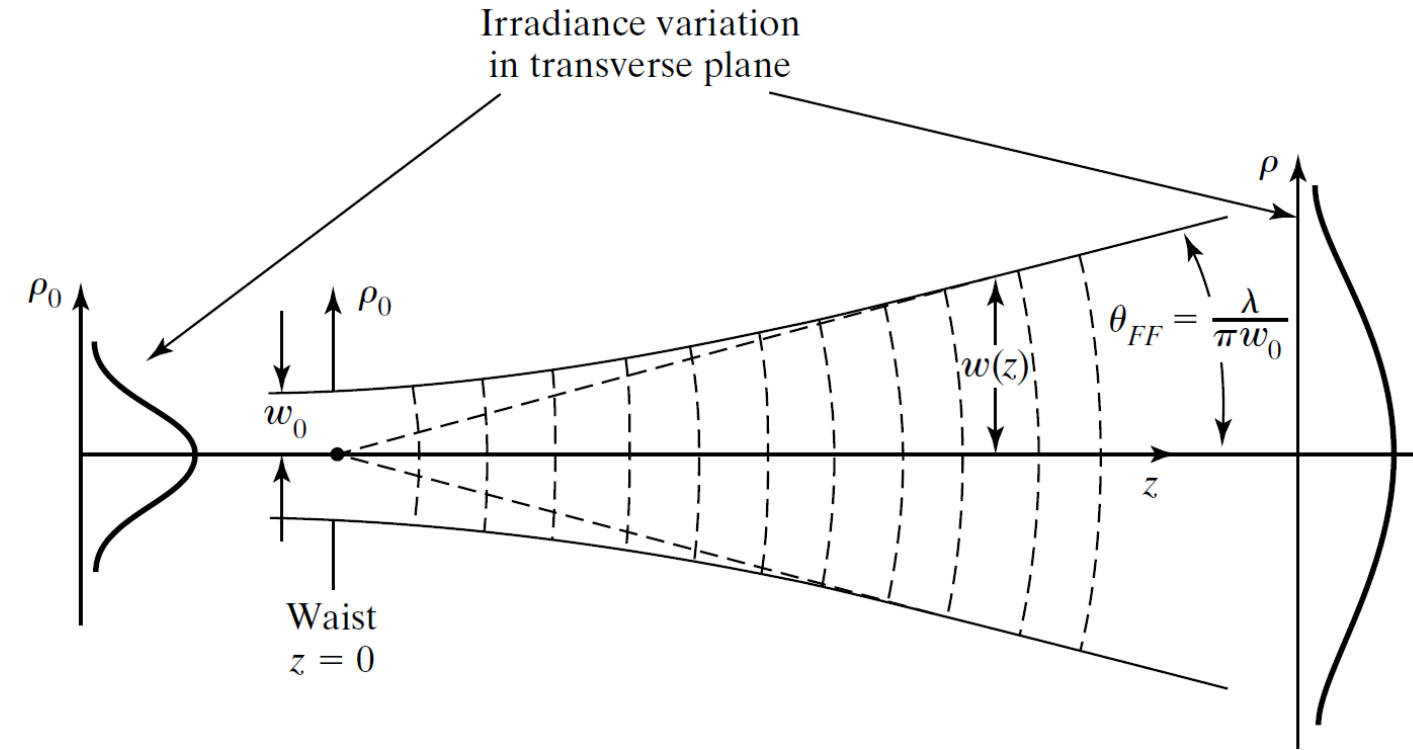


Figure 2 Gaussian spherical beam propagating in the z -direction. The spot size $w(z)$ at the beam waist ($z = 0$) is defined as w_0 . The half-angle beam divergence $\theta_{FF} = \lambda/(\pi w_0)$ is valid only in the far field. Note the change in transverse irradiance as the beam propagates to the right.

$$I(\rho, z) = I_0 \left(\frac{w_0}{w(z)} \right)^2 e^{-2\rho^2/w^2(z)}$$

$$I_0 = I(\rho = 0, z = 0)$$

$$I(\rho = 0, z) = I_0 \left(\frac{z_0^2}{z^2 + z_0^2} \right)$$

Axial irradiance

$$w(z = 0) = w_0$$

$$w(z = z_0) = \sqrt{2}w_0$$

$$w(z \gg z_0) = \frac{w_0 z}{z_0} = \frac{\lambda}{\pi w_0} z$$

Spot size at the beam waist

Spot size at $z = z_0$

Spot size in the far field ($z \gg z_0$)

$$\frac{I(\rho, z)}{I(\rho = 0, z)} = e^{-2\rho^2/w^2(z)}$$

Gaussian Laser Beams

Example 11.4

Write the beam waist w_0 in terms of the *f-number*, defined to be the ratio of z to the beam diameter $2w(z)$ far from the beam waist.

Solution: Far away from the beam waist (i.e. $z \gg z_0$) the laser beam expands along a cone. That is, its diameter increases in proportion to distance.

$$w(z) = w_0 \sqrt{1 + z^2/z_0^2} \rightarrow w_0 z/z_0$$

The cone angle is parameterized by the f-number, the ratio of the cone height to its base:

$$f^{\#} \equiv \lim_{z \rightarrow \pm\infty} \frac{z}{2w(z)} = \frac{z}{2w_0 z/z_0} = \frac{z_0}{2w_0}$$

Substitution of (11.44) into this expression yields

$$w_0 = \frac{2\lambda f^{\#}}{\pi} \quad (11.46)$$

Equation (11.46) gives a convenient way to predict the size of a laser focus. One calculates the f-number by dividing the diameter of the beam far from the focus into the distance from the focus. In practice you may be very surprised at how poorly a beam may focus in comparison with the theoretical prediction (due to aberrations). It is always good practice to directly measure your focus if its size is important to an experiment.

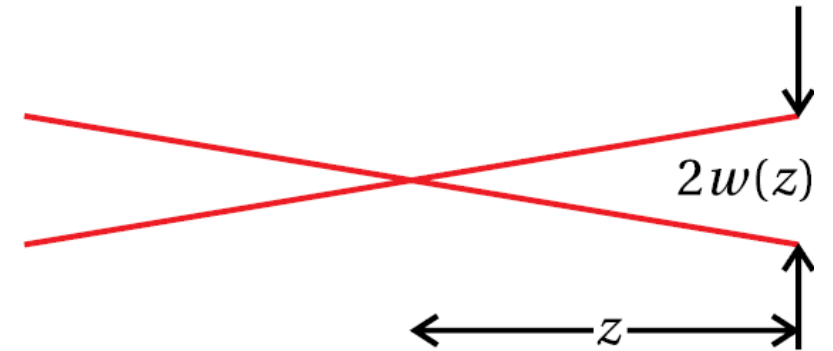


Figure 11.18